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KPI-Based Approach for Business Process Improvement

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Abstract

Business Process Management is a contemporary approach with the main purpose of deploying, executing and continuously optimizing the different types of business processes and thus improving the agility of the organization. In this paper, we propose an approach to business process improvement based on key performance indicators during the Business Process Management lifecycle. We started by highlighting the life cycle of key performance indicators. Then our proposal was at meta-modeling level, adding performance indicators as concepts of the Business Process Model and Notation meta-model. This brought us to a new extension of this meta-model based on key performance indicators. The particular case of the Scrum agile development process is then considered as a case study. In this context, key performance indicators are proposed for this process and are classified according to the elements of the Scrum method. The proposed approach was tested by taking the example of a company using ScrumWise tool for conducting scrum developments. The developed prototype allowed a performance evaluation of the Scrum process through key performance indicators and a set of recommendations were proposed to help managing tasks and improving the adopted process.

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Keywords: Key Performance Indicator (KPI), Business Process Management, Business Process Model and Notation meta-model, SCRUM method, ScrumWise.

1. Introduction

Business Process Management (BPM) has become one of the most widely used approaches in the company. It allows to design, monitor, configure, analyze and evaluate processes [8]. An important aspect in business process lifecycle is the evaluation of business processes performance [9]. Performance requirements on business processes are specified

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as KPIs with target values which must be reached in a certain period. For improving business processes, evaluation is a necessary step. This is due to the fact that evaluation provides different measurements that indicate whether company goals are successfully achieved or not. Thus appears the usefulness of KPIs. Moreover, the BPMN allows companies to better understand their internal business processes using a graphical notation and communicate them in a standard way. This notation is described by its meta-model, for which we propose an extension including KPI-related concepts. The remainder of this paper is organized as follows. In section 1 we present our proposed meta-model. Then In Section 2 we present a case study concerning Scrum development process. Then in Section 3 we propose an organized list of KPIs referring to our case study and finally in Section 4 to present an overview of a prototype we developed, based on the Meta-model of BPMN that facilitates and manages the KPI lifecycle in this context of Scrum development. In Section 5 we draw conclusions from our analysis and outline our future work.

2. The proposed meta-model

2.1. The design of the BPMN meta-model

The BPMN standard was developed by the Object Management Group (OMG). This standard allows companies to better understand their internal business processes using a graphical notation and communicates them in a standard way [1]. The fundamental role of BPMN is to provide a notation comprehensible to all business users from business analysts who do the first draft of business processes to the technical developers responsible for implementing the technology needed to the execution and the users of the company who will manage and monitor these processes. Therefore, BPMN comes to fill the gap between business process modeling and their implementation [1]. Many meta-models exist in literature such as the BPMN meta-model diagram for the definition of business processes by [2] and Diagram meta-model OMG BPMN by [3].

2.2. BPMN Meta-model with KPI extension

Since we are following a BPM approach that promotes continuous improvement of business processes, we have suggested an extension to a simplified version of the BPMN Meta-model. We will improve it by adding KPI-related concepts. In Figure 1, we have added to the BPMN Meta-model concepts related to KPIs that includes: Global Process KPI, Event KPI, Lane KPI, Activity KPI, Sub Process KPI and Task KPI. The concepts of (process, activity, event, pool, sub-Process, task, and lane) already exist in the pre-cited Meta-models.

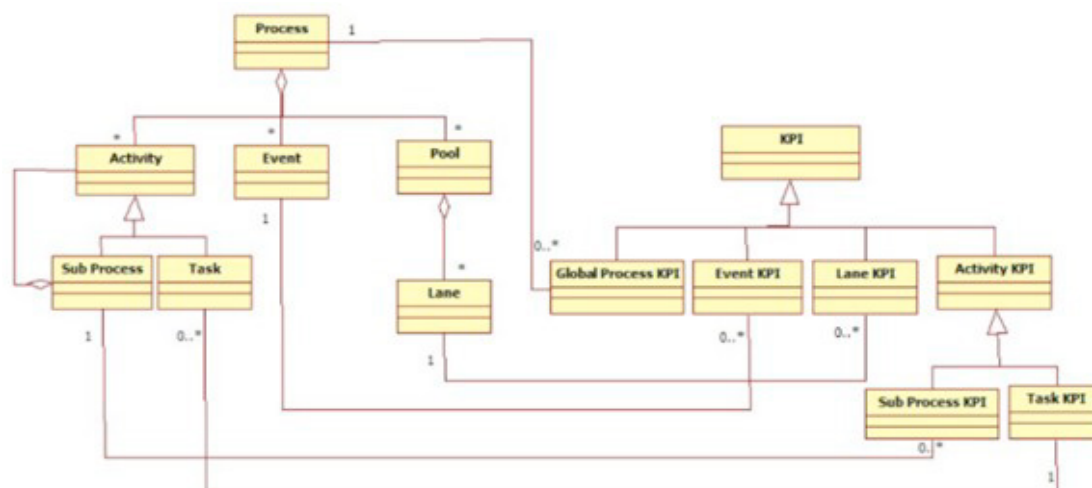


Figure 1: Meta-model BPMN design with KPI extension

In this extension, we have enriched the BPMN Meta-model mainly with a KPI class (super class). Generally, a KPI is the measure of an activity expressed by a numerical value. It also has a description and threshold values. The KPI super class is specialized into all types of KPIs of a business process such as: Global Process KPI, Event KPI, Lane

KPI, Activity KPI, Sub Process KPI, and Task KPI.

Depending on the value taken by the type attribute, we end up with the following classes:

- Global Process KPI: measures the overall performance of a business process.
- Event KPI: measures the performance of an event.
- Lane KPI: measures the performance of a lane.
- Activity KPI: measures the performance of an Activity.
- Sub Process KPI: measures the performance of a Sub Process.
- Task KPI: measures the performance of a Task.

3. Case study

Our case study concerns the agile development process of scrum method with the purpose of improving the management of software development processes through BPM and KPIs.

3.1. SCRUM Overview

Scrum is a framework in which people can solve complex adaptation problems while productively and creatively delivering products of the greatest possible value.

3.2. Adoption of KPIs for SCRUM performance measurement

Most software development projects do not respect time, budget, quality and / or customer expectations. Studies have shown the need for a significant improvement in these processes and practices of this type of project. In this context, to reduce the risks of this type of problem we have focused on the application of Business Process Management and KPIs to improve the management of software development processes of the Scrum method.

3.3. Proposed Classification of KPIs

3.3.1. Classification of KPIs

We have identified a list of KPIs that can be measured during the realization of a scrum project and classified them according to the Scrum concepts as:

- Event related : considered events are sprints, daily scrum, sprint planning, sprint review and retrospective sprint
- Artifact related : considered artifacts are product backlog, sprint backlog, project, user story, task, team

3.3.2. Examples of the KPIs proposed

Examples of KPIs associated to User Story are the following:

- Number of tasks associated with a user story.
- Number of tasks completed in a user story.

Examples of KPIs associated to Task are the following:

- Estimated number of hours for a task.
- The actual number of hours of a task.

Examples of KPIs associated to Sprint are the following:

- Number of user stories completed at the end of the sprint compared to the total number of user stories in the sprint backlog.
- Number of user stories not started at the end of the sprint compared to the total number of user stories in the sprint backlog.

Examples of KPIs associated to Daily Scrum are the following:

- Number of tasks performed compared to the tasks planned for the previous day.
- Number of outstanding and unfinished tasks in relation to the tasks of the day previous.

- Number of problems encountered in completing the tasks of the previous day.

We have proposed to a SCRUM team (Scrum Master and development team) to fill out a table work of the min and max thresholds including the list of key performance indicators (KPIs) of our approach.

The objectives we aim to achieve:

- Evaluate and measure the performance of the development team, project performance, productivity, timeliness, efficiency, project progress, flexibility, etc.
- Propose recommendations in order to open areas for improvement in the company and in group work, which guarantees better quality for the client, the user and the developer.

4. Developed Prototype

4.1. Identification of the features of our prototype

Main features are presented in figure 2.

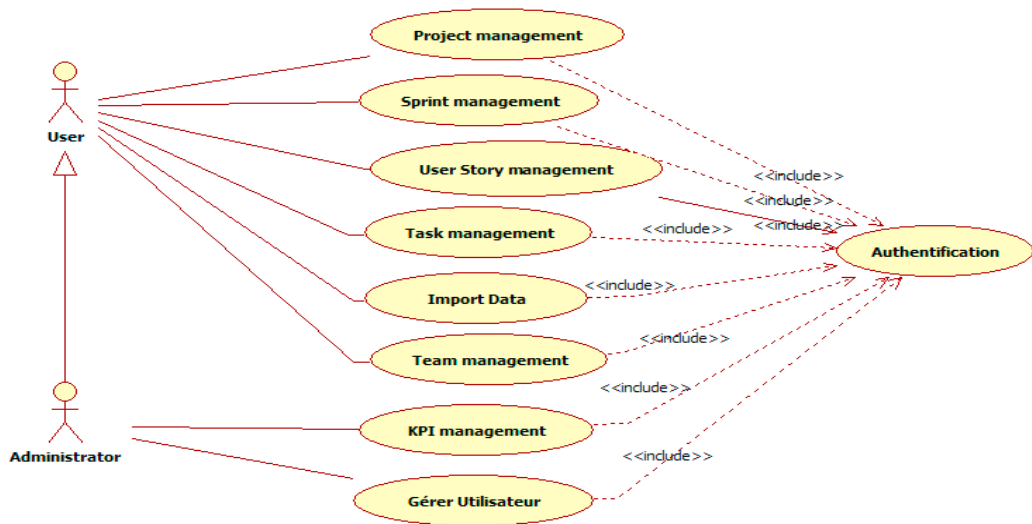


Figure 2: the prototype use case diagram

4.2. Class diagram of our prototype

The process depends on the progress of the development project within a computer company that uses the Scrum method. For this, we find the following classes in figure 3 that make up the Scrum method: a sprint, a user story, a task, a team and a project.

For each project:

- A project is composed of 1 or more sprints.
- A sprint is composed of 1 or more user stories.
- A user story is composed of 1 or more tasks.
- Each project is carried out by 1 or more teams.

The evaluation of each project is done through KPIs, the types of KPIs are:

- Project KPI, Sprint KPI, User Story KPI, Task KPI, Team KPI

Measurements of KPI values allow project evaluation as well as the proposal recommendations that would help the user to improve the performance of his projects and make the right decisions for their situation. We adopt this structure to take into account the elements of the scrum method and KPIs. For this we proposed a prototype that allows users of Scrum teams on the one hand to evaluate the performance of the project and on the other hand to gain from recommendations to helping in better managing tasks and improving work process.

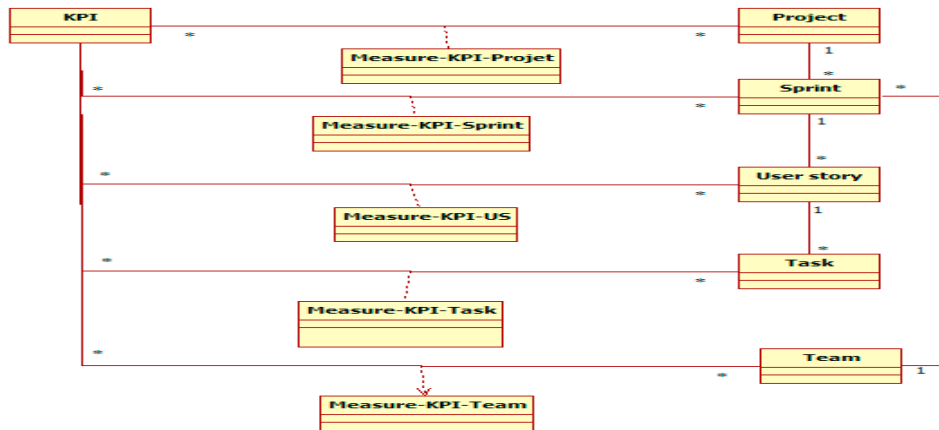


Figure 3: Prototype Class diagram

Some KPIs are entered manually and others are automatically pulled from a csv file (generated from the scrum wise). Then, the user of the application is able to add the necessary data of the project for which he wants to evaluate performance through the corresponding that KPIs.

The user of the application is able to add the necessary data of the sprints of the project that he wants to evaluate: validate the data or cancel or import the files related to the given sprint from the Scrum Wise.

4.3. Developed Interfaces

At the prototype level in figure 4 and 5, the values are compared with the thresholds and messages are displayed to the users informing them about the obtained level of performance.

In figure 6, the user of the application can also directly import the necessary data related to the user stories of his project.

↳ Mesurer Performance > KPIs

KPI User Story

Projet

Sprint

User Story

Nom KPI

Valeur KPI

Figure 4: Measurement of User story KPI

s KPI User Story

Projet

Sprint

User Story

Nom KPI

Valeur KPI

ation KPI User Story

Nom KPI

KPI Message

Figure 5: Evaluation of User story KPI

The screenshot shows a web interface for importing data. At the top, there is a button labeled 'Importer Données'. Below it, there are three input fields. The first field is labeled 'Projet' and contains the text 'Yellow Korner'. The second field is labeled 'Sprint' and contains the text 'Sprint1'. The third field is labeled 'Fichier' and has a dropdown menu with the option 'Choisir un fichier' and a file named 'Scrumwise...12-20.csv' selected.

Figure 6: Import Data.

5. Validation of the solution

For validating our solution and the developed prototype, we took the real case of a development project developed by the company "SoftyGo" that was conducted by the Scrum method and ScrumWise [7] as a management tool. This project called "Yellow korner" aims to develop a website of a French art photography publishing. Results were useful to the company. In fact, the results produced by the prototype made it possible to evaluate the performance of the "Yellow korner" project. Indeed, global KPIs, relating to different sprints, different user stories and even tasks could be measured, compared with their thresholds and recommendations were displayed whenever the performance was low. It has been considered by company staff that this performance measure and these recommendations are very useful for improving performance during a project or for future projects.

6. Conclusion

Our solution was at the BPM lifecycle in which we highlighted the lifecycle of KPIs. Then, our proposal was at the BPMN meta-model level by adding Performance Indicators (KPIs) as an extension of the meta-model. As a result, we have obtained an extended meta-model from a simplified version of the OMG abstract model. For highlighting the benefits realized by this work, we took Scrum process as a case study, for which we proposed performance indicators (KPIs) and we developed a prototype that allows to evaluate the performance of this process through performance indicators (KPIs) and able to track the workflow in a project by giving recommendations for improving subsequent tasks. For validating our solution, we chose to work on an example within the company "SoftyGo" for which we were able to run our prototype and obtain results considered useful for the staff of the company. Finally, let's note that our work remains open to a number of improvements and perspectives such as the following:

- Improve our solution by applying quality practices and lead to a more general version of our prototype in order to evaluate the performance of business processes.
- Apply more reliable techniques to determine thresholds for KPIs.

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